

Adafruit modular control panel

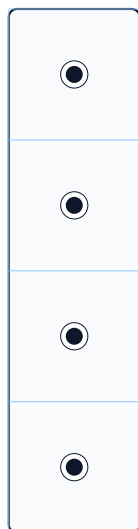
A 3D-printable 1.00 in (25.4 mm) grid. Print strips, screw them together from below, stand the boards off the grid with M3 standoffs, and build one solid control surface. Featured parts: NeoSlider 5295, quad rotary 5752, ANO 6310, rotary 5880 / 4991, plus Adafruit displays.

Locked grid

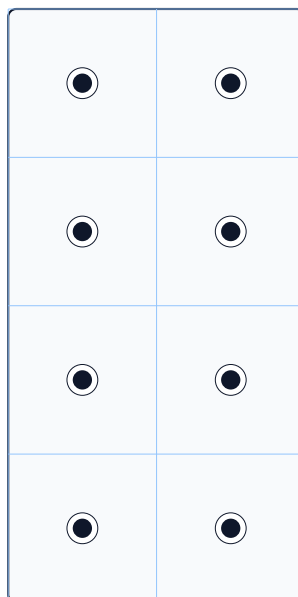
Item	Value	Why
Pitch	1.00 in / 25.4 mm	Matches the 1x1 hole request. Rotary 4991 is a 1.00 in square PCB.
Strip width	1 to 5 units (25.4 to 127.0 mm)	1-wide for sliders and single knobs. 2-wide for ANO. 3 to 5 wide for LCD/TFT.
Strip length	4 units / 101.6 mm	Fits a 3.00 in NeoSlider or quad rotary with margin. Tile length up to 5 strips = 508 mm.
Max assembly	5 wide x 5 long = 127.0 x 508.0 mm	Printable as separate 4 in strips. A 5x4 plate is 127 x 102 mm and fits a Bambu 256 mm bed.
Grid hole	3.3 mm through, 6.5 mm x 1.8 mm countersink on the bottom	M3 screw from below into a female-female standoff. Head sits flush under the panel.
Panel thickness	4.0 mm	Stiff enough as a 1-wide strip. Print countersink on the bed, no supports.
Board standoff	M3 FF 11–12 mm	STEMMA QT connectors are on the bottom of the QT boards. Cables run in the gap.
Board screws	M2.5 into the Adafruit PCB holes	Adafruit plated holes are 2.5 mm. Use the printed adapter so 2.5 mm and 1 in holes do not have to coincide.
Join	Tongue + groove on +X/+Y and -X/-Y, plus a 2-hole join-bar under the seam	Screws from below clamp two strips onto the same standoffs. The 1 in pitch stays continuous across the joint.

The 1 in holes sit at the centre of each cell, not at the corners. A 1-wide strip is 25.4 mm and has one column of holes. Adafruit board holes are 2.5 mm and sit about 0.10 in in from the PCB corners, so they do not land on the 1 in grid. Print the adapter plate for that board: M3 holes on the grid, M2.5 holes on the PCB, standoffs in between.

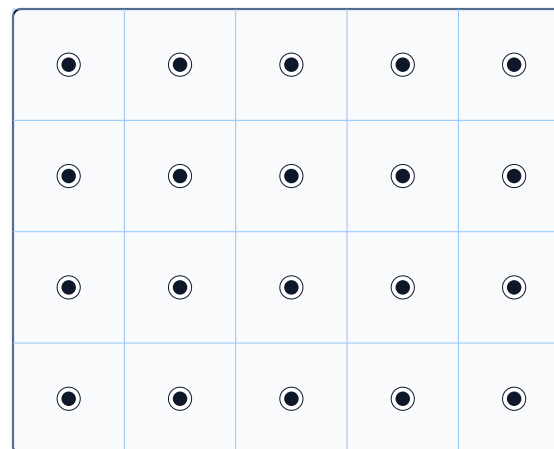
Print these STLs



1x4 1x4 units 25.4 x 101.6 mm



2x4 2x4 units 50.8 x 101.6 mm



5x4 LCD 5x4 units 127.0 x 101.6 mm

Files in cad/stl/: strip_1x4, strip_2x4, strip_3x4, strip_4x4, strip_5x4, join_bar, adapter_4991 (also 5880), adapter_5295, adapter_5752, adapter_6310, and matching faceplates. Print PLA or PETG, 0.20 mm layer, 3 walls, 30% gyroid, no supports, countersink on the bed. Source: cad/grid-panel.scad.

Featured devices — measured from Adafruit Eagle PCB files

Hole coordinates are millimetres from the origin stated in the row. Cut-throughs are for an optional faceplate that sits above the knobs. The base grid itself is a solid 1 in hole plate with no device cutouts.

ID	Device	PCB mm	Mounting holes	Cut-through	Grid
6310	Adafruit ANO Rotary Navigation Encoder to I2C Adapter (encoder soldered) PCB 1.40 x 1.60 in. Needs a 2-wide strip. Same PCB as 5740 without the encoder.	35.56 x 40.64 mm	M2.5 / 2.5 mm dia (15.24, 17.78); (-15.24, 17.78); (15.24, -17.78); (-15.24, -17.78) origin board center	1 x dia 22.0 mm (ANO click-wheel through a faceplate)	2 wide x 2 long (50.8 x 50.8 mm) print 2x4 strip
5880	Adafruit I2C Stemma QT Rotary Encoder Breakout (encoder soldered) Identical PCB to 4991 with PEC11 soldered. Shaft at center. Same 1.00 in footprint.	25.4 x 25.4 mm	M2.5 / 2.5 mm dia (2.54, 2.54); (22.86, 2.54); (2.54, 22.86); (22.86, 22.86) origin board lower-left	1 x dia 7.5 mm (PEC11 bushing (faceplate))	1 wide x 1 long (25.4 x 25.4 mm) print 1x4 strip
4991	Adafruit I2C Stemma QT Rotary Encoder Breakout (no encoder) Exactly 1.00 in square. Encoder at 45 degrees, shaft at board center. Same PCB as 5880.	25.4 x 25.4 mm	M2.5 / 2.5 mm dia (2.54, 2.54); (22.86, 2.54); (2.54, 22.86); (22.86, 22.86) origin board lower-left	1 x dia 7.5 mm (PEC11 bushing (faceplate))	1 wide x 1 long (25.4 x 25.4 mm) print 1x4 strip
5752	Adafruit I2C Quad Rotary Encoder Breakout Same PCB outline as NeoSlider. Four PEC11 footprints on 19.05 mm (0.75 in) pitch. Encoders not included.	76.2 x 21.59 mm	M2.5 / 2.5 mm dia (19.05, 8.26); (-19.05, 8.26); (19.05, -8.26); (-19.05, -8.26) origin board center	4 x dia 7.5 mm (PEC11 bushing (faceplate))	1 wide x 3 long (25.4 x 76.2 mm) print 1x4 strip
5295	Adafruit NeoSlider I2C QT Slide Potentiometer PCB is 3.00 x 0.85 in. Shop height 21.5 mm includes the slider knob. STEMMA QT is on the bottom; use 10-12 mm M3 standoffs.	76.2 x 21.59 mm	M2.5 / 2.5 mm dia (19.05, 8.26); (-19.05, 8.26); (19.05, -8.26); (-19.05, -8.26) origin board center	slot 75.0 x 4.0 mm (75 mm slide pot travel (faceplate))	1 wide x 3 long (25.4 x 76.2 mm) print 1x4 strip

How each featured board sits

Board	Strip	Adapter	Above the board
5295 NeoSlider	1-wide x 4 long	adapter_5295.stl — 4 M3 on the grid, 4 M2.5 at ± 19.05 / ± 8.255 mm (board rotated so 3 in runs along the strip)	faceplate_5295.stl slot 75 x 4 mm, or leave open
5752 Quad rotary	1-wide x 4 long	adapter_5752.stl — same PCB outline as the NeoSlider. Four PEC11 centres at $x = \pm 9.525$, ± 28.575 mm	faceplate_5752.stl four 7.5 mm holes. PEC11 bushing is ~7 mm; 7.5 mm is the panel hole.
4991 / 5880 rotary	1-wide x 4 long (uses 2 cells)	adapter_4991.stl is 1x2 so two M3 standoffs stop rotation. Four M2.5 at 2.54 mm from each corner of the 1.00 in PCB.	faceplate_4991.stl one 7.5 mm hole over the shaft at board centre.
6310 ANO	2-wide x 2 long (or a 2x4 strip)	adapter_6310.stl — 4 M3, 4 M2.5 at ± 15.24 / ± 17.78 mm. 4.0 mm encoder posts are cleared, do not put screws in them.	faceplate_6310.stl 22 mm hole for the click wheel.

Adafruit displays — shop-page sizes and strip count

Adafruit category LCDs & Displays was searched (TFT, OLED, character LCD, eInk, HDMI backpacks). Rows below have a published millimetre size. Hole spans are used when the shop page lists them; otherwise mount with the 1 in grid and an adapter, or 2.5 mm holes 2.5 mm in from the PCB corners (Adafruit default, not verified per display). Window cut-throughs use the published viewing area, or the PCB minus 2 mm if the view size is not listed (marked inferred). A display that needs more than 5 strips wide or more than 5 lengths of 4 in does not fit the modular max; it is marked overflow.

ID	Device	Kind	Size mm	Mount	Cut-through	Strips W x L
181	Standard LCD 16x2 + extras - white on blue	character-lcd	36 x 80.6	not published on shop page	window 32.0 x 76.6 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 4
4196	Adafruit 1.54" Monochrome eInk / ePaper Display with SRAM - 200x200 with SSD1681 and EYESPI	eink	43.5 x 43	not published on shop page	window 39.5 x 39.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 2
4868	Adafruit 1.54" Tri-Color eInk / ePaper Display with SRAM - 200x200 with SSD1681 and EYESPI	eink	43.5 x 43	not published on shop page	window 39.5 x 39.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 2
4195	Adafruit 2.13" Monochrome eInk / ePaper Display FeatherWing - 250x122 Monochrome with SSD1680	eink	61.3 x 40.2	not published on shop page	window 57.3 x 36.2 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 3
4197	Adafruit 2.13" Monochrome eInk / ePaper Display with SRAM - 250x122 Monochrome with SSD1680	eink	61.5 x 39.3	not published on shop page	window 57.5 x 35.3 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 3
4777	Adafruit 2.9" Grayscale eInk / ePaper Display FeatherWing - SSD1680 - Mono / 4 Grayscale	eink	79.5 x 38	M2.5 / 2.5 mm dia (-37.15, -21.00); (37.15, -21.00); (-37.15, 21.00); (37.15, 21.00) origin board center (span from shop page)	window 75.5 x 34.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 4
1028	Adafruit 2.9" Red/Black/White eInk Display Breakout - THINK INK - SSD1680 Chipset	eink	80 x 47	not published on shop page	window 76.0 x 43.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 4
4778	Adafruit 2.9" Tri-Color eInk / ePaper Display FeatherWing - SSD1680 - Red Black White	eink	79.5 x 38	M2.5 / 2.5 mm dia (-37.15, -21.00); (37.15, -21.00); (-37.15, 21.00); (37.15, 21.00) origin board center (span from shop page)	window 75.5 x 34.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 4
6418	Adafruit E-Ink Bonnet for Raspberry Pi - 24-pin E-Paper Displays	eink	65 x 30.7	not published on shop page	window 61.0 x 26.7 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 3
4098	Adafruit 2.7" Tri-Color eInk / ePaper Display with SRAM - Red Black White	eink	72.5 x 56.7	not published on shop page	window 68.5 x 52.7 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 3
4800	Adafruit MagTag - 2.9" Grayscale E-Ink WiFi Display - 2025 Edition with SSD1680	eink	80 x 53.5	not published on shop page	window 76.0 x 49.5 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 4

ID	Device	Kind	Size mm	Mount	Cut-through	Strips W x L
5023	Bluetooth® eInk Display Clock with Temperature Humidity Sensor - LYWSD02	eink	110 x 55	not published on shop page	window 106.0 x 51.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 5
2900	Adafruit FeatherWing OLED - 128x32 OLED Add-on For Feather	oled	22.9 x 50.9	not published on shop page	window 18.9 x 46.9 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	1 x 3
4650	Adafruit FeatherWing OLED - 128x64 OLED Add-on For Feather - STEMMA QT / Qwiic	oled	22.9 x 50.9	not published on shop page	window 18.9 x 46.9 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	1 x 3
3527	Adafruit PiOLED - 128x32 Monochrome OLED Add-on for Raspberry Pi	oled	35 x 20	not published on shop page	window 31.0 x 16.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	1 x 2
3045	Assembled Adafruit FeatherWing OLED - 128x32 OLED Add-on For Feather	oled	22.9 x 50.9	not published on shop page	window 18.9 x 46.9 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	1 x 3
6313	Assembled Adafruit FeatherWing OLED - 128x64 OLED Add-on - STEMMA QT / Qwiic	oled	52 x 23	not published on shop page	window 48.0 x 19.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	1 x 3
4440	Monochrome 0.91" 128x32 I2C OLED Display - STEMMA QT / Qwiic	oled	20 x 35	not published on shop page	window 7.0 x 25.0 mm (display window (faceplate))	1 x 2
661	Monochrome 128x32 SPI OLED graphic display	oled	23 x 33	M2 / 2.0 mm dia (-7.50, -12.00); (7.50, -12.00); (-7.50, 12.00); (7.50, 12.00) origin board center (span from shop page)	window 7.0 x 25.0 mm (display window (faceplate))	1 x 2
3531	Adafruit 128x64 OLED Bonnet for Raspberry Pi	oled	65.3 x 30.7	not published on shop page	window 61.3 x 26.7 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 3
5297	Adafruit Monochrome 1.12" 128x128 OLED Graphic Display - STEMMA QT / Qwiic	oled	40.5 x 30.5	not published on shop page	window 36.5 x 26.5 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 2
326	Monochrome 0.96" 128x64 OLED Graphic Display - STEMMA QT	oled	29.2 x 26.7	not published on shop page	window 26.6 x 19.0 mm (display window (faceplate))	2 x 2
938	Monochrome 1.3" 128x64 OLED graphic display - STEMMA QT / Qwiic	oled	35.6 x 33	M2.5 / 2.5 mm dia (-15.25, -14.00); (15.25, -14.00); (-15.25, 14.00); (15.25, 14.00) origin board center (span from shop page)	window 34.5 x 23.0 mm (display window (faceplate))	2 x 2
2675	Monochrome 2.3" 128x32 OLED Graphic Display Module Kit	oled	66.5 x 35	not published on shop page	window 62.5 x 31.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 3

ID	Device	Kind	Size mm	Mount	Cut-through	Strips W x L
1673	OLED Breakout Board - 16-bit Color 1.27" w/microSD holder - EYESPI Connector	oled	35.8 x 43	not published on shop page	window 31.8 x 39.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 2
5100	Adafruit MACROPAD RP2040 Bare Bones - 3x4 Keys + Encoder + OLED	oled	104 x 59.5	not published on shop page	window 100.0 x 55.5 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 5
5345	Adafruit ESP32-S2 Reverse TFT Feather	tft	52 x 23	not published on shop page	window 48.0 x 19.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	1 x 3
5300	Adafruit ESP32-S2 TFT Feather - 4MB Flash, 2MB PSRAM, STEMMA QT	tft	52.4 x 22.8	not published on shop page	window 48.4 x 18.8 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	1 x 3
6303	Adafruit ESP32-S3 Reverse TFT with w.FL Antenna	tft	52.2 x 22.8	not published on shop page	window 48.2 x 18.8 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	1 x 3
5483	Adafruit ESP32-S3 TFT Feather - 4MB Flash, 2MB PSRAM, STEMMA QT	tft	52.5 x 22.8	not published on shop page	window 48.5 x 18.8 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	1 x 3
3321	Adafruit Mini Color TFT with Joystick FeatherWing	tft	50 x 23	not published on shop page	window 46.0 x 19.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	1 x 2
4393	Adafruit Mini PiTFT - 135x240 Color TFT Add-on for Raspberry Pi	tft	38.6 x 24.5	not published on shop page	window 34.6 x 20.5 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	1 x 2
618	1.8" SPI TFT display, 160x128 18-bit color - ST7735R driver	tft	46.6 x 34.5	not published on shop page	window 42.6 x 30.5 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 2
4311	2.0" 320x240 Color IPS TFT Display with microSD Card Breakout - ST7789 EYESPI	tft	59.2 x 35.5	not published on shop page	window 55.2 x 31.5 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 3
2770	2.8" TFT Display - 240x320 with Capacitive Touchscreen	tft	69 x 50	not published on shop page	window 65.0 x 46.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 3
3533	Adafruit 0.96" 160x80 Color TFT Display w/ MicroSD Card Breakout - ST7735	tft	32 x 28	not published on shop page	window 28.0 x 24.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 2
4383	Adafruit 1.14" 240x135 Color TFT Display + MicroSD Card Breakout - ST7789	tft	32 x 27.8	not published on shop page	window 28.0 x 23.8 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 2

ID	Device	Kind	Size mm	Mount	Cut-through	Strips W x L
4313	Adafruit 1.3" 240x240 Wide Angle TFT LCD Display with MicroSD - ST7789	tft	35.8 x 35.8	M2.5 / 2.5 mm dia (-16.50, -15.00); (16.50, -15.00); (-16.50, 15.00); (16.50, 15.00) origin board center (span from shop page)	window 26.0 x 26.0 mm (display window (faceplate))	2 x 2
4506	Adafruit 1.3" Color TFT Bonnet for Raspberry Pi - 240x240 TFT + Joystick Add-on	tft	65.5 x 30.6	not published on shop page	window 61.5 x 26.6 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 3
5393	Adafruit 1.47" 320x172 Round Rectangle Color IPS TFT Display - ST7789	tft	37.3 x 30.5	M2.5 / 2.5 mm dia (-15.24, -12.70); (15.24, -12.70); (-15.24, 12.70); (15.24, 12.70) origin board center (span from shop page)	window 34.0 x 19.4 mm (display window (faceplate))	2 x 2
3787	Adafruit 1.54" 240x240 Wide Angle TFT LCD Display with MicroSD - ST7789 with EYESPI Connector	tft	43.7 x 41.8	M2.5 / 2.5 mm dia (-19.05, -17.78); (19.05, -17.78); (-19.05, 17.78); (19.05, 17.78) origin board center (span from shop page)	window 32.0 x 31.0 mm (display window (faceplate))	2 x 2
5206	Adafruit 1.69" 280x240 Round Rectangle Color IPS TFT Display - ST7789	tft	45.8 x 36.8	M2.5 / 2.5 mm dia (-19.05, -15.24); (19.05, -15.24); (-19.05, 15.24); (19.05, 15.24) origin board center (span from shop page)	window 38.0 x 30.0 mm (display window (faceplate))	2 x 2
5394	Adafruit 1.9" 320x170 Color IPS TFT Display - ST7789	tft	57.2 x 36.6	M2.5 / 2.5 mm dia (-26.00, -15.85); (26.00, -15.85); (-26.00, 15.85); (26.00, 15.85) origin board center (span from shop page)	window 53.2 x 32.6 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 3
4484	Adafruit Mini PiTFT 1.3" - 240x240 TFT Add-on for Raspberry Pi	tft	38.4 x 31.5	not published on shop page	window 34.4 x 27.5 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	2 x 2
2090	2.8" TFT LCD with Cap Touch Breakout Board w/MicroSD Socket - EYESPI Connector	tft	81.3 x 62.5	M2.5 / 2.5 mm dia (-38.10, -28.60); (38.10, -28.60); (-38.10, 28.60); (38.10, 28.60) origin board center (span from shop page)	window 69.6 x 45.2 mm (display window (faceplate))	3 x 4
1743	3.2" TFT LCD with Touchscreen Breakout Board w/MicroSD Socket - ILI9341	tft	90 x 60	not published on shop page	window 86.0 x 56.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 4
2050	3.5" TFT 320x480 + Touchscreen Breakout Board w/MicroSD Socket - HX8357D	tft	56 x 97	not published on shop page	window 56.0 x 85.0 mm (display window (faceplate))	3 x 4
802	Adafruit 1.8" Color TFT Shield w/microSD and Joystick - v 2	tft	68.5 x 53	not published on shop page	window 64.5 x 49.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 3

ID	Device	Kind	Size mm	Mount	Cut-through	Strips W x L
2478	Adafruit 2.4" TFT LCD with Touchscreen Breakout w/MicroSD Socket - ILI9341	tft	53 x 60	not published on shop page	window 49.0 x 56.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 3
2423	Adafruit PiTFT Plus 320x240 2.8" TFT + Capacitive Touchscreen	tft	56 x 85	not published on shop page	window 52.0 x 81.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 4
5872	Adafruit TFT FeatherWing - 3.5" 480x320 Capacitive Touchscreen - STEMMMA QT / Qwiic	tft	89.3 x 66	not published on shop page	window 85.3 x 62.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 4
3651	Adafruit TFT FeatherWing - 3.5" 480x320 Touchscreen for Feathers - V2 with TSC2007	tft	84 x 65	not published on shop page	window 80.0 x 61.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 4
4367	Circuit Playground TFT Gizmo - Bolt-on Display + Audio Amplifier - JST PH 2mm	tft	53.3 x 53.3	not published on shop page	window 49.3 x 49.3 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 3
2616	PiTFT Plus 320x240 3.2" TFT + Resistive Touchscreen	tft	56 x 85	not published on shop page	window 52.0 x 81.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 4
2298	PiTFT Plus Assembled 320x240 2.8" TFT + Resistive Touchscreen	tft	56 x 85	not published on shop page	window 52.0 x 81.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 4
5852	Round RGB 666 TTL TFT Display - 2.8" 480x480 - No Touch - TL028WVC01-B1621A	tft	102 x 73	not published on shop page	window 98.0 x 69.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 5
3315	TFT FeatherWing - 2.4" 320x240 Touchscreen For All Feathers - V2	tft	65 x 53	not published on shop page	window 61.0 x 49.0 mm inferred (display window, inferred 2 mm inset from PCB (not measured))	3 x 3

Assembly

1. Print one strip per column. A mixer of knobs uses 1-wide strips. A 2.8 in TFT uses a 4-wide (101.6 mm) or a 5-wide (127 mm) plate. A 3.5 in TFT is 4-wide by 3 long — print a 4x4 and leave one row empty, or tile a second 4x4 on the length.
2. Lay strips on a table, countersink down. Slide +X tongues into the next strip's -X grooves. Length joints use +Y into -Y. The 1 in holes line up across the seam.
3. Put a join_bar under each seam so its two holes match one hole from each strip. From below, run an M3 screw through the join-bar, through both countersinks, into an 11–12 mm female-female standoff on top. That screw both joins the strips and starts a standoff.
4. Fill the remaining 1 in holes the same way wherever you will put a board. Empty holes can stay open or take a short M3 screw and nut so the panel stays a solid sheet.
5. Print the adapter for the board. Screw the adapter onto the standoffs with M3. Screw the Adafruit PCB onto the adapter with M2.5. STEMMA QT cables run in the 11–12 mm gap.
6. Optional: print the faceplate, stand it off above the knobs, and let shafts / the slider / the display window come through the cut-outs.
7. Hardware per cell you use: 1 x M3x8 to 12 mm pan or CSK from below, 1 x M3 FF standoff 11–12 mm, 1 x M3x6 into the adapter. Per board: 4 x M2.5x6 mm into the PCB.

Print settings that were used to generate the STLs

OpenSCAD nightly, cad/grid-panel.scad, PART= each name. Countersink is on the -Z face. Bambu / any FDM: 0.4 mm nozzle, 0.20 mm layer, 3 walls, 30% gyroid, 40 mm/s outer wall, no supports, no brim required on the 4x4 and 5x4, a 3 mm brim on the 1x4 strip so it does not peel. PETG if the panel will sit in a warm rack; PLA is fine for a desk. Do not heat-set inserts in the 4 mm plate — the through hole plus a metal standoff is the joint.

Sources

PCB holes and outlines: Adafruit Eagle files Adafruit-NeoSlider-PCB, Adafruit-I2C-QT-Rotary-Encoder-PCB, Adafruit-I2C-Quad-Rotary-Encoder-Breakout-PCB, Adafruit-ANO-Rotary-Navigation-Encoder-to-I2C. Shop dimensions: <https://www.adafruit.com/api/products> plus each product technical-details block. PEC11 panel hole 7.5 mm is the common bushing clearance, not a dimension on the Adafruit PCB.